

Quantum-Assisted Few-Shot and Cross-Domain Chest X-Ray Classification

Pipeline 4 retained in BOTH forms: ViT/CheXzero (Ours) and RadJEPA (Ours)

Item	Current setting
Source domain	NIH ChestX-ray
Target domain	CheXpert
Pathologies	Atelectasis, Cardiomegaly, Consolidation, Edema, Pleural Effusion
Primary metrics	Per-pathology binary AUROC / macro AUROC; AUPRC; balanced accuracy; F1
Pipeline 4 - ViT	Pipeline 4 FQ-1 (Ours, ViT/CheXzero): 6q, 3 reuploads, 18-D reducer, weighted density memory, exact Bures
Pipeline 4 - RadJEPA	Pipeline 4 FQ-1 (Ours, RadJEPA): RadJEPA last-2 adaptation + same 6q density/fidelity framework
Status	Development comparison; not a quantum-advantage claim

Important: The earlier ViT/CheXzero Pipeline 4 results are retained. The new RadJEPA run is added as a separate Ours row; it does NOT replace the earlier Pipeline 4 row.

1. Headline Results - Both Ours Rows

Method	1-shot AUROC	5-shot AUROC	1-shot AUPRC	5-shot AUPRC	Interpretation
Stage-A classical base - ViT/CheXzero	88.48%	88.82%	93.23%	93.50%	Earlier strong base
Pipeline 4 FQ-1 (Ours, ViT/CheXzero)	88.95%	88.82%	93.23%	93.50%	+0.48 pp at 1-shot; tie at 5-shot
Stage-A classical base - RadJEPA	88.83%	88.76%	92.76%	92.68%	Latest RadJEPA base
Pipeline 4 FQ-1 (Ours, RadJEPA)	88.32%	86.67%	92.75%	92.58%	-0.51 pp at 1-shot; -2.09 pp at 5-shot

The ViT/CheXzero Pipeline 4 remains the earlier positive 1-shot development result. The RadJEPA version is a separate experiment and currently underperforms its own Stage-A base. Both should be shown together rather than replacing one with the other.

2. Pipelines 1 and 2 - Retained

Method	1-shot AUROC	5-shot AUROC	10-shot AUROC	Interpretation
Plain classical k-shot	54.74%	60.31%	63.68%	Weak baseline
Pipeline 1 quantum E5	55.44%	60.72%	63.88%	Small gain over plain head
Classical cosine retrieval	59.29%	62.90%	64.84%	Best of these three

Method	1-shot AUROC	5-shot AUROC	Role
QuantFSL base	71.02%	72.31%	Main classical base
Pipeline 2 trainable 4-qubit fidelity	71.02%	72.34%	Main quantum branch
PCA4 cosine	71.01%	72.42%	Strong matched classical
PCA4 RBF	71.02%	72.39%	Strong matched classical
Matched MLP4	70.99%	72.34%	Parameter-matched trainable
RFF16	71.02%	72.38%	Explicit feature control

3. Direct Comparison of Pipelines 1, 2, and BOTH Pipeline 4 Variants

Property	Pipeline 1	Pipeline 2	Pipeline 4 (Ours, ViT/CheXzero)	Pipeline 4 (Ours, RadJEPA)
Core idea	Mixed density state from patches	Whole-image CheXzero + 4q residual	6q density memory + exact Bures rerank	RadJEPA + 6q density memory + exact Bures rerank
1-shot AUROC	55.44%	71.02%	88.95%	88.32%
5-shot AUROC	60.72%	72.34%	88.82%	86.67%
Increment vs own base	Small vs plain head	Essentially tied	~+0.48 pp at 1-shot; ~0 at 5-shot	-0.51 pp at 1-shot; -2.09 pp at 5-shot
Main limitation	Salvage/provenance issue	Matched controls tie	FAST: 3 support seeds, 1 model seed	FAST + RadJEPA target-pretraining exposure

4. Cross-Dataset Generalisation: NIH -> CheXpert

Method	1-shot AUROC	1-shot AUPRC	1-shot Bal.Acc	1-shot F1	5-shot AUROC	5-shot AUPRC	5-shot Bal.Acc	5-shot F1
Stage-A Base - ViT/CheXzero	88.48	93.23	-	-	88.82	93.50	-	-
Pipeline 4 FQ-1 (Ours, ViT/CheXzero)	88.95	93.23	80.14	82.53	88.82	93.50	79.44	82.32
Stage-A Base - RadJEPA	88.83	92.76	80.48	77.25	88.76	92.68	77.78	85.47
Pipeline 4 FQ-1 (Ours, RadJEPA)	88.32	92.75	79.39	78.19	86.67	92.58	75.87	84.31
Parameter-Matched MLP - RadJEPA	53.62	67.79	57.99	54.02	53.62	67.79	57.99	54.02
Classical RBF - RadJEPA	52.67	66.61	51.16	46.65	58.42	71.00	57.23	52.76
Random Fourier Features - RadJEPA	53.44	65.80	49.81	49.63	58.24	70.60	57.76	57.35
Mean Causal Patch - RadJEPA	52.51	66.56	49.95	51.71	58.03	71.12	56.93	68.22

A dash means the exact corresponding metric was not reported in the earlier ViT/CheXzero update table; it is left blank rather than invented.

5. Pipeline 4 Representation / Fidelity Ablations - BOTH Runs

Run	Ablation	Variant	Shot	AUROC	AUPRC	Bal.Acc	F1
ViT/CheXzero (Ours)	Exact rerank	Profile coarse only	1	88.18	93.16	80.02	81.78
ViT/CheXzero (Ours)	Exact rerank	Profile + exact Bures	1	88.95	93.23	80.14	82.53
ViT/CheXzero (Ours)	Exact rerank	Profile coarse only	5	88.82	93.51	79.50	82.41
ViT/CheXzero (Ours)	Exact rerank	Profile + exact Bures	5	88.82	93.50	79.44	82.32
ViT/CheXzero (Ours)	Set encoding	Mean patch vector	1	42.83	66.86	50.85	43.09
ViT/CheXzero (Ours)	Set encoding	Weighted density operator	1	88.95	93.23	80.14	82.53
ViT/CheXzero (Ours)	Set encoding	Mean patch vector	5	56.60	77.63	60.28	61.57
ViT/CheXzero (Ours)	Set encoding	Weighted density operator	5	88.82	93.50	79.44	82.32
ViT/CheXzero (Ours)	Weighting	Pathology logit proxy	1	87.34	93.01	79.11	83.39
ViT/CheXzero (Ours)	Weighting	Uniform	1	86.17	92.98	76.96	80.02
ViT/CheXzero (Ours)	Weighting	Pathology logit proxy	5	88.86	93.53	79.32	82.18
ViT/CheXzero (Ours)	Weighting	Uniform	5	88.81	93.49	79.35	82.17
RadJEPA (Ours)	Exact rerank	Profile coarse only	1	88.79	92.79	79.36	78.10
RadJEPA (Ours)	Exact rerank	Profile + exact Bures	1	88.32	92.75	79.39	78.19
RadJEPA (Ours)	Exact rerank	Profile coarse only	5	87.91	92.66	76.00	84.44
RadJEPA (Ours)	Exact rerank	Profile + exact Bures	5	86.67	92.58	75.87	84.31
RadJEPA (Ours)	Set encoding	Mean patch vector	1	52.51	66.56	49.95	51.71
RadJEPA (Ours)	Set encoding	Weighted density operator	1	88.32	92.75	79.39	78.19
RadJEPA (Ours)	Set encoding	Mean patch vector	5	58.03	71.12	56.93	68.22
RadJEPA (Ours)	Set encoding	Weighted density operator	5	86.67	92.58	75.87	84.31
RadJEPA (Ours)	Weighting	Pathology logit proxy	1	87.81	92.49	78.27	78.68
RadJEPA (Ours)	Weighting	Uniform	1	88.40	92.54	78.45	79.24
RadJEPA (Ours)	Weighting	Pathology logit proxy	5	86.21	92.63	75.68	84.33
RadJEPA (Ours)	Weighting	Uniform	5	87.37	92.69	75.86	84.49

6. Literature Context - BOTH Ours Rows

Method	Venue / Year	Backbone	1-shot Acc	1-shot AUC	5-shot Acc	5-shot AUC	Mean novel AUC
ProtoNet + ViT	Diagnostics, 2025	ViT-B/16*	61.42 +/- 0.53	72.11 +/- 0.61	74.33 +/- 0.41	82.67 +/- 0.44	72.14
ProtoMed	IVC, 2025	ResNet-12	63.08 +/- 0.47	73.84 +/- 0.58	76.19 +/- 0.38	83.91 +/- 0.42	73.87
MetaChest	VCIA, 2026	ResNet-12	64.17 +/- 0.51	74.36 +/- 0.55	77.04 +/- 0.37	84.53 +/- 0.41	74.55
LP++ / CLIP-CXR	MICCAI, 2024	BioViL-T*	65.93 +/- 0.49	76.02 +/- 0.52	79.11 +/- 0.35	86.78 +/- 0.38	76.43
Expert-Guided FSL	IEEE JBHI, 2026	ResNet-12	63.74 +/- 0.52	74.65 +/- 0.59	77.42 +/- 0.39	85.21 +/- 0.40	75.26
DC-ProtoNet	IEEE JBHI, 2024	ResNet-50*	64.89 +/- 0.48	75.17 +/- 0.53	78.07 +/- 0.36	85.74 +/- 0.39	75.78
WME-Net (ResNet-12)	Supplied table	ResNet-12	67.84 +/- 0.44	78.59 +/- 0.47	81.37 +/- 0.31	88.24 +/- 0.34	78.59
Pipeline 4 FQ-1 (Ours, ViT/CheXzero)	Current development run	ViT/CheXzero + 6q density	N/A*	88.95	N/A*	88.82	N/A*
Pipeline 4 FQ-1 (Ours, RadJEPA)	Current development run	RadJEPA + 6q density	N/A*	88.32	N/A*	86.67	N/A*

Table II. NIH -> CheXpert Cross-Dataset Comparison

Method	Setting	1-shot Acc	5-shot Acc	1-shot AUC	5-shot AUC
ProtoNet + ViT	NIH -> CXP	58.24	71.38	68.93	80.14
LP++ / CLIP-CXR	NIH -> CXP	61.07	74.83	71.44	83.59
ProtoMed	NIH -> CXP	57.93	70.12	67.28	79.47
MetaChest	NIH -> CXP	59.61	72.04	69.53	80.98
DC-ProtoNet	NIH -> CXP	60.17	73.55	70.29	82.11
WME-Net (ResNet-12)	NIH -> CXP	62.39	75.61	72.81	84.07
Pipeline 4 FQ-1 (Ours, ViT/CheXzero)	NIH -> CXP	N/A*	N/A*	88.95	88.82
Pipeline 4 FQ-1 (Ours, RadJEPA)	NIH -> CXP	N/A*	N/A*	88.32	86.67

Table III. Numerical AUC Gap vs Strongest Supplied Cross-Dataset Row

Comparison	1-shot AUC	5-shot AUC	Delta vs WME-Net
WME-Net (ResNet-12)	72.81	84.07	Reference
Pipeline 4 FQ-1 (Ours, ViT/CheXzero)	88.95	88.82	+16.14 pp; +4.75 pp
Pipeline 4 FQ-1 (Ours, RadJEPA)	88.32	86.67	+15.51 pp; +2.60 pp

Comparability warning: Both Ours rows use a different multi-label patient-level evaluation than the supplied few-shot literature rows. The RadJEPA run also reports target-domain pretraining exposure. These gaps are contextual only, not direct SOTA evidence.

7. RadJEPA-Specific Additional Ablations

Reference-bank size K

Run	Reference K	Trainable References	Source Recall@10	Best
ViT/CheXzero (Ours)	64	64	Not separately reported in supplied update	Primary setting
RadJEPA (Ours)	16	16	61.67%	
RadJEPA (Ours)	32	32	61.67%	
RadJEPA (Ours)	64	64	64.17%	Primary setting
RadJEPA (Ours)	128	128	65.00%	Best recall

Efficiency

Run	Trainable Params	Qubits	Re-upload Blocks	Reference K	Source Val AUROC	Source Val AUPRC	Recall@10
ViT/CheXzero (Ours)	Not separately reported	6	3	64	-	-	-
RadJEPA (Ours)	7,061	6	3	64	61.56%	30.68%	64.17%

8. Final Supervisor-Level Takeaway

Keep both results in the manuscript draft. **Pipeline 4 FQ-1 (Ours, ViT/CheXzero)** is the earlier development configuration with 88.95% / 88.82% macro AUROC and a small positive 1-shot increment over its Stage-A base. **Pipeline 4 FQ-1 (Ours, RadJEPA)** is the newer backbone experiment with 88.32% / 86.67% macro AUROC, but its Stage-A RadJEPA base is higher. These two rows answer different representation questions and should be presented as separate Ours variants, not as replacement results.